

EXPERIMENT 8

Query with Sub Query & Correlated Query

AIM

To perform **Sub-Query and Correlated Sub-Query** operations on the Hospital database.

The manual defines a subquery as a query nested inside another query, while a correlated subquery uses data from the outer query.

DESCRIPTION:

SUBQUERY

A MySQL subquery is a query nested within another query such as SELECT, INSERT, UPDATE or DELETE. In addition, a MySQL subquery can be nested inside another subquery.

A MySQL subquery is called an inner query while the query that contains the subquery is called an outer query. A subquery can be used anywhere that expression is used and must be closed in parentheses.

CORRELATED QUERY:

A correlated subquery is a subquery that uses the data from the outer query. In other words, a correlated subquery depends on the outer query. A correlated subquery is evaluated once for each row in the outer query.

QUESTIONS

1. Find the patients whose age is greater than the average age of all patients.
2. Find the doctors whose date of birth is earlier than the average date of birth of all doctors.

4. List the patients whose age is greater than the average age of patients of their respective gender.

1)

Patient_ID	Patient_Name	Age	Gender	Disease	Phone_No
P003	Rahul	30	M	Asthma	9876543212
P004	Sneha	28	F	Migraine	9876543213
NULL	NULL	NULL	NULL	NULL	NULL

```
267 • SELECT *
268 FROM Doctor
269 WHERE TIMESTAMPDIFF(YEAR, DOB, CURDATE()) >
270 (SELECT AVG(TIMESTAMPDIFF(YEAR, DOB, CURDATE()))
271 FROM Doctor);
272
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content							
Doctor_ID	Doctor_Name	Gender	DOB	Specialization	Phone_No	Department	
D001	Ravi	M	1985-04-12	Cardiology	9876500001	Emergency	
D003	Kumar	M	1982-11-15	Orthopedics	9876500003	Orthopedics	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	

3)

```
274 • SELECT *
275 FROM Doctor
276 WHERE DOB = (SELECT MIN(DOB) FROM Doctor);
277
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content

	Doctor_ID	Doctor_Name	Gender	DOB	Specialization	Phone_No	Department
▶	D003	Kumar	M	1982-11-15	Orthopedics	9876500003	Orthopedics
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4)

```
279 • SELECT *
280 FROM Patient P
281 WHERE Age > (
282     SELECT AVG(Age)
283     FROM Patient
284     WHERE Gender = P.Gender
285 );
286
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	Patient_ID	Patient_Name	Age	Gender	Disease	Phone_No
	P003	Rahul	30	M	Asthma	9876543212
	P004	Sneha	28	F	Migraine	9876543213
*	NULL	NULL	NULL	NULL	NULL	NULL

RESULT:

The records from the tables are displayed using Sub-Query and Correlated Sub-Query.